

Assignment 1 Report

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Facial Affect Recognition using CNN Baselines

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1. Network Details (10 pts)

ResNet50 Model

- **Architecture:** Pretrained ResNet50 backbone. Final fully connected layer replaced with three heads:
  - Expression classification (8 classes, softmax)
  - Valence regression (1 output, tanh)
  - Arousal regression (1 output, tanh)
- **Parameters:** ∼23 million
- **Training Settings:** Optimizer: Adam, Learning Rate: 1e-4, Batch Size: 32, Epochs: 10
- **Rationale:** ResNet50 is a strong backbone with deep feature extraction ability, commonly used in image recognition tasks.

EfficientNet-B0 Model

- **Architecture:** Pretrained EfficientNet-B0 backbone, final FC replaced with same three heads.
- **Parameters:** ∼5.3 million
- **Training Settings:** Same as ResNet50
- **Rationale:** EfficientNet scales efficiently and provides strong accuracy with fewer parameters, acting as a lightweight comparison.

2. Comparison of Multiple Baselines (10 pts)

Table 1: Performance comparison of baselines (after augmentation)

Metric	ResNet50	EfficientNet-B0
Accuracy	0.3983	**0.4350**
F1 Score	0.3972	**0.4336**
Cohen's Kappa	0.3129	**0.3546**
Valence RMSE	0.4177	**0.3960**
Valence Corr	**0.5726**	0.5546
Arousal RMSE	0.3804	**0.3706**
Arousal Corr	**0.3934**	0.3678

Observation: EfficientNet-B0 outperformed ResNet50 in most metrics after augmentation, especially in classification accuracy and regression RMSE. ResNet50 maintained slightly higher correlation for valence and arousal. Overall, EfficientNet-B0 appears more robust with augmentation, while ResNet50 retains stronger feature correlations.

3. Transfer Learning Used (5 pts)

Both models were initialized with ImageNet pretrained weights. Transfer learning helped speed up convergence and improved performance by reusing low-level features (edges, textures, shapes). The task-specific heads (classification and regression) were trained from scratch.

4. Training Graphs (5 pts)

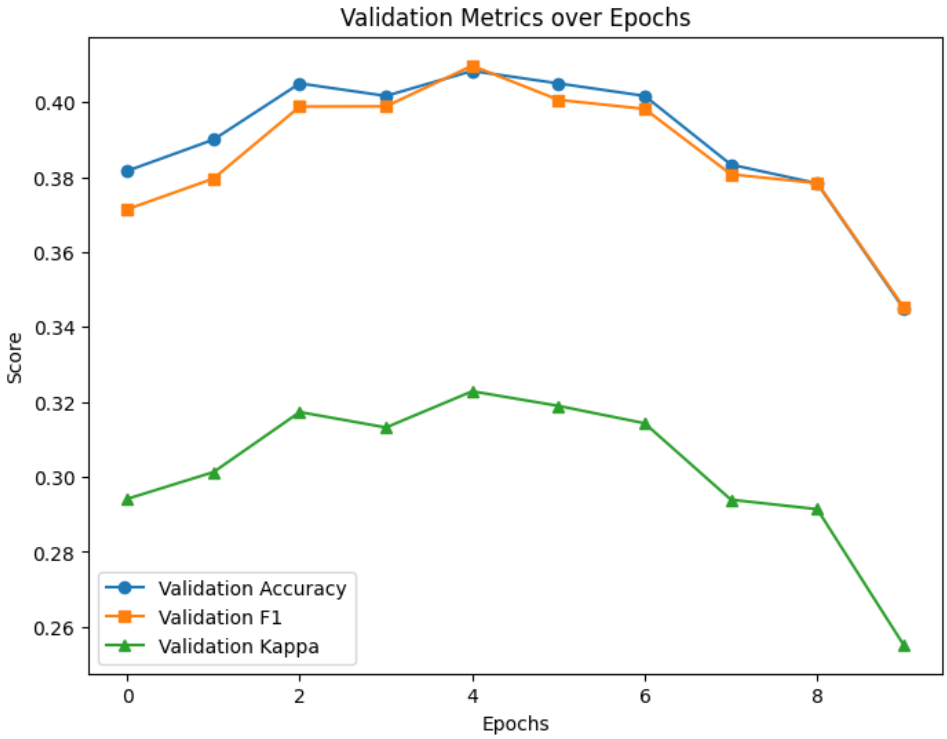


Figure 1: Validation Metrics over Epochs (ResNet50)

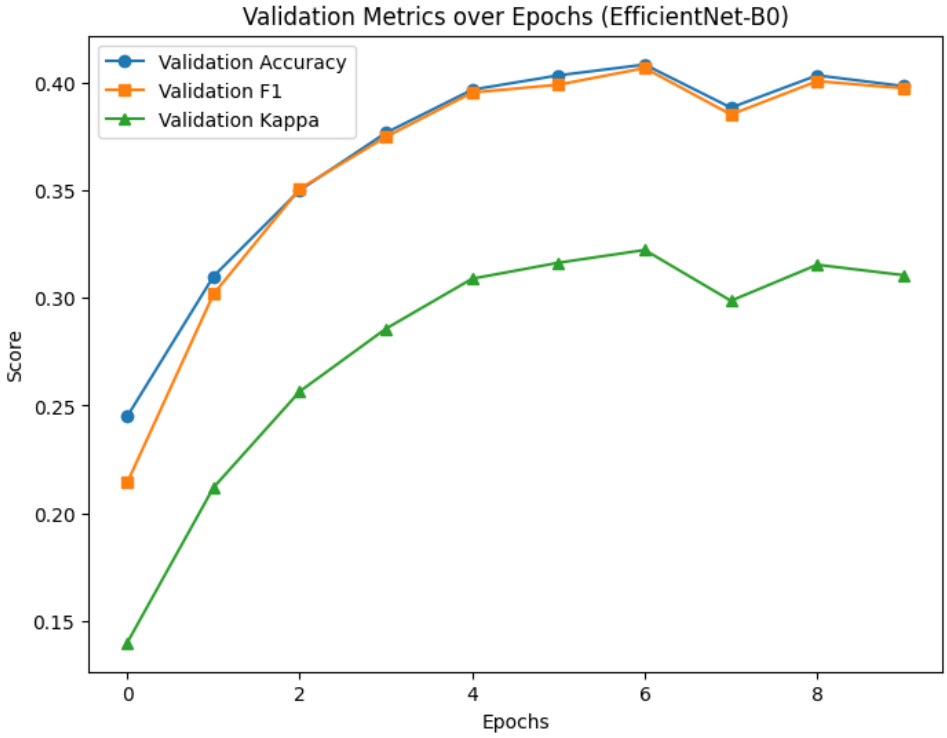


Figure 2: Validation Metrics over Epochs (EfficientNet-B0)

5. Performance Measures & Discussion (15 pts)

Metrics Used

- **Accuracy & F1 Score:** Measure correctness in categorical prediction. F1 is especially useful under class imbalance.
- **Cohen's Kappa:** Measures agreement beyond chance, robust under skewed class distributions.
- **RMSE (Root Mean Squared Error):** Standard error metric for regression, penalizes larger mistakes.
- **Pearson Correlation (CORR):** Evaluates linear agreement and trend consistency between predictions and ground truth.

Other Continuous Metrics

- **SAGR (Sign Agreement):** Focuses on whether the predicted sign of valence/arousal matches ground truth, reflecting emotional polarity (positive vs negative, active vs passive).
- **CCC (Concordance Correlation Coefficient):** Combines correlation and mean error, widely regarded as a balanced measure in affective computing.

Discussion

For affective computing systems in the wild, it is not enough to just minimize RMSE or maximize correlation.

- RMSE ensures predictions are numerically close.
- CORR ensures the predicted trend matches real affect variation.
- SAGR ensures emotional polarity (positive vs negative, calm vs excited) is captured correctly, which is essential for interaction systems.
- CCC balances both correlation and bias, making it the most reliable real-world measure.

Thus, CCC and SAGR are the most suitable metrics for practical deployment, ensuring predictions are both directionally correct and numerically reliable.

6. Correctly and Incorrectly Classified Images (4 pts)

Examples of model outputs:

- **Correctly classified:** Happy $\rightarrow$ Happy, Neutral $\rightarrow$ Neutral
- **Incorrectly classified:** Sad $\rightarrow$ Neutral, Fear $\rightarrow$ Surprise

Sample Predictions

We show a few examples of correctly and incorrectly classified images from the test set.

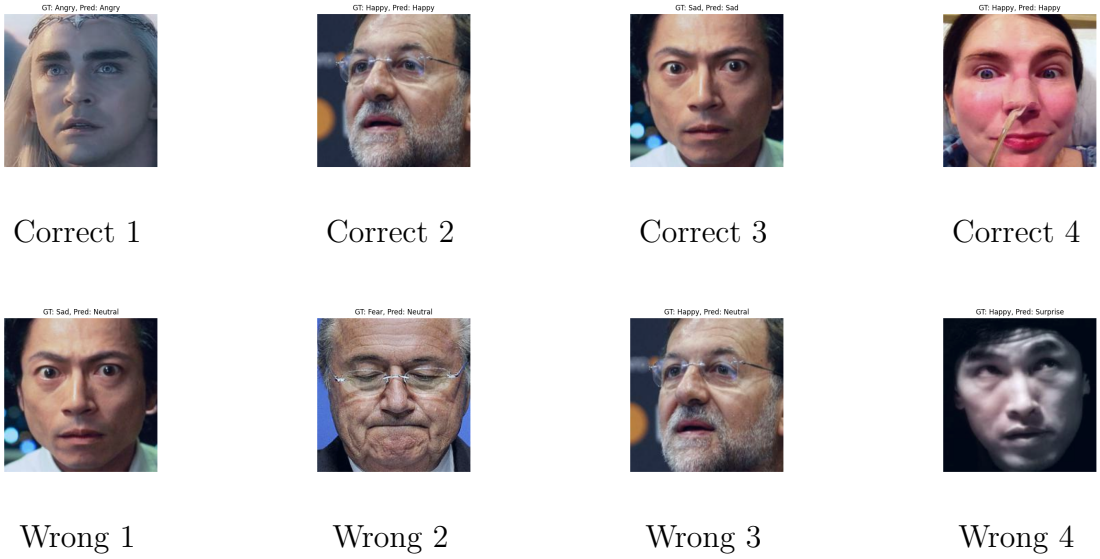


Figure 3: Examples of correctly and incorrectly classified facial expression images.

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